LAPORAN PRAKTIKUM

POSTTEST (1)

ALGORITMA PEMROGRAMAN LANJUT

Disusun oleh:

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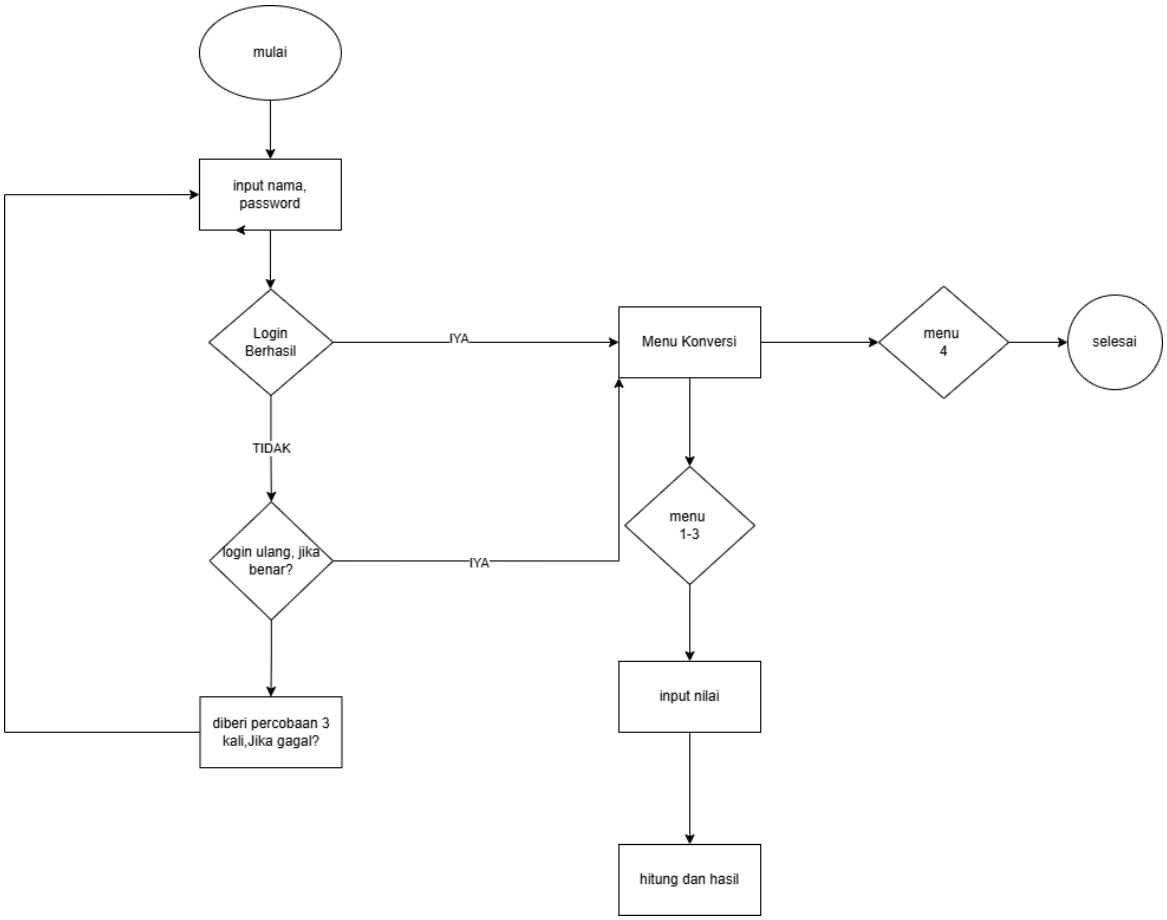
PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

SAMARINDA

2025

1. Flowchart



Gambar 2.1 Flowchart

2. Deskripsi Singkat Program

Program ini adalah program konversi satuan panjang yang diawali dengan sistem login. User harus memasukkan nama dan password (3 digit terakhir NIM) dengan maksimal 3 kali percobaan. Jika gagal sebanyak 3 kali, program akan berhenti.

3. Source code

```cpp
#include <iostream>
using namespace std;

int main() {
    string nama, password;
    int kesempatan = 3;

    // LOGIN
    while (kesempatan--) {
        cout << "=== LOGIN ===\n";
        cout << "Nama     : ";
        cin >> nama;
        cout << "Password : ";
        cin >> password;

        if (nama == "Dika" && password == "130") {
            cout << "Login berhasil!\n";
            break;
        } else {
            cout << "Salah! ";
            if (kesempatan > 0)
                cout << "Sisa kesempatan: " << kesempatan << "\n\n";
            else {
                cout << "\nAnda gagal login 3 kali.\n";
                return 0;
            }
        }
    }

    int pilih;
    double nilai;

    // MENU
    do {
        cout << "\n=== MENU KONVERSI ===\n";
        cout << "1. Meter ke Km & Cm\n";
        cout << "2. Km ke Meter & Cm\n";
        cout << "3. Cm ke Meter & Km\n";
        cout << "4. Keluar\n";
        cout << "Pilihan: ";
        cin >> pilih;

        if (pilih == 4) {
            cout << "Program selesai.\n";
            break;
        }

        if (pilih < 1 || pilih > 4) {
            cout << "Menu tidak ada!\n";
            continue;
        }

        cout << "Masukkan nilai: ";
        cin >> nilai;

        if (pilih == 1) {
            cout << nilai << " m = "
                 << nilai / 1000 << " km dan "
                 << nilai * 100 << " cm\n";
        }
        else if (pilih == 2) {
            cout << nilai << " km = "
                 << nilai * 1000 << " m dan "
                 << nilai * 100000 << " cm\n";
        }
        else if (pilih == 3) {
            cout << nilai << " cm = "
                 << nilai / 100 << " m dan "
                 << nilai / 100000 << " km\n";
        }

    } while (true);

    return 0;
}
```

Gambar 2.2 Source Code

```
=== LOGIN ===
Nama     : Dika
Password : 130
Login berhasil!
```

Gambar 2.3 Login

```
=== MENU KONVERSI ===
1. Meter ke Km & Cm
2. Km ke Meter & Cm
3. Cm ke Meter & Km
4. Keluar
Pilihan:
```

Gambar 2.4 Menu Konversi

```
Masukkan nilai: 10
10 m = 0.01 km dan 1000 cm

=== MENU KONVERSI ===
1. Meter ke Km & Cm
2. Km ke Meter & Cm
3. Cm ke Meter & Km
4. Keluar
Pilihan:
```

Gambar 2.5 Menu 1

```
Masukkan nilai: 100
100 km = 100000 m dan 1e+007 cm

=== MENU KONVERSI ===
1. Meter ke Km & Cm
2. Km ke Meter & Cm
3. Cm ke Meter & Km
4. Keluar
Pilihan:
```

Gambar 2.6 Menu 2

```
Masukkan nilai: 1000
1000 cm = 10 m dan 0.01 km

=== MENU KONVERSI ===
1. Meter ke Km & Cm
2. Km ke Meter & Cm
3. Cm ke Meter & Km
4. Keluar
Pilihan:
```

Gambar 2.7 Menu 3

```
Program selesai.
PS C:\Users\USER\Documents\Praktikum-APL>
```

Gambar 2.8 Menu 4

Langkah Langkah GIT

GIT Add

```
PS C:\Users\USER\Documents\Praktikum-APL> git add .
```

GIT Commit

```
PS C:\Users\USER\Documents\Praktikum-APL> git commit -m "Finish Posttest 1"
[main 7c24110] Finish Posttest 1
 5 files changed, 81 insertions(+)
 create mode 100644 posttest apl/2509106130-RadhikaAdhityaArifin-PT-1.pdf
 rename 2509106130-RadhikaAdhityaArifin-PT1.cpp => posttest apl/2509106130-RadhikaAdhityaArifin-PT1.cpp (100%)
 create mode 100644 posttest apl/2509106130-RadhikaAdhityaArifin-PT1.exe
 create mode 100644 posttest apl/tempCodeRunnerFile.cpp
 create mode 100644 posttest apl/tempCodeRunnerFile.exe
```

GIT Push

```
PS C:\Users\USER\Documents\Praktikum-APL> git push
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (1/1), done.
Writing objects: 100% (2/2), 240 bytes | 240.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/adhityaradhika3/Praktikum-APL.git
   7c24110..761f324  main -> main
PS C:\Users\USER\Documents\Praktikum-APL>
```